

Decoding the Lipid-Protein Interface of Membrane Transporters: From Allosteric Dynamics to Structural Pharmacology

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Membrane transporters, including the ATP-Binding Cassette (ABC) and Solute Carrier superfamilies, are critical gatekeepers governing cellular homeostasis, local pharmacokinetics and physiological states. While structural biology has made available a myriad of static structures, understanding their structural pharmacology requires investigating their dynamic behaviour within a realistic lipid bilayer membrane.

We here leverage multi-microsecond all-atom molecular dynamics simulations to decipher the complex interplay between membrane, conformational plasticity and structural dynamics across several membrane transporters, namely ABCB4, MRP1, and OAT1. Our computational insights revealed that the surrounding lipid bilayer membrane does not act as a passive container but rather specific lipid-mediated allosteric communications modulate the signal transduction between substrate binding pockets and functional domains (e.g., ABC nucleotide binding domains or SLC charge-relay system). We also bridge these fundamental biophysical mechanisms to translational approaches. For instance, we investigated the impact of single mutations on the structural dynamics of defective intracellularly-retained class II ABCB4 variants. Furthermore, we also scale these transporter functions into a biomimetic proximal tubule-on-chip model to evaluate the transcellular transporter-mediated transport of xenobiotics. Together, our multi-scale approach enables an understanding of local PK at the molecular and cellular level.